VRAJ VAGHELA

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EDUCATIONS

Sardar Vallabhbhai National Institute of Technology (SVNIT) | Surat, Gujarat — Aug 2023 - Present
Bachelor of Technology in Computer Science and Engineering — *CGPA : 8.76/10.0*
- Expected graduation in **May 2027**

Netaji Model School | Vadodara, Gujarat — Jun 2022 - Apr 2023
Senior Secondary (GHSEB, Science) — *CGPA : 82%*
- Graduated in **April 2023** with **82%**

PROJECTS

Enfinity (Briefly): Distraction-Free Adaptive Reading Platform | Netlify — MMM 2023
Developer
- Crafted a real-time adaptive reading platform using a proprietary **Comprehension Friction Score** (CFS) engine combining Dale-Chall text difficulty, observed reading speed (WPM), and scroll regression rate to detect per-paragraph reader struggle
- Engineered a micro-engagement telemetry system (**Zustand**, **IntersectionObserver**) tracking dwell time, scroll regression, and mouse hesitation (>800ms) to trigger contextual NLP interventions across 3 adaptive reading modes (Auto, Ask Me, Manual)
- Developed an NLP adaptation pipeline (compromise.js) with a multi-tier intervention strategy: acronym expansion, ESL cognate swapping via **LibreTranslate**, and 3-layer definition fallback
- Integrated **Google Gemini 2.5 Flash** for AI-powered summarization, **Supabase** authentication backed by 4 schema tables, and local-first persistence via **Dexie.js** with 5-second debounced saves and 150ms scroll recovery

Technologies / Tools Used : *React, TypeScript, Vite, Zustand, Express.js, NLP*

Retail Stockout Prediction & Inventory Transfer System | Github — MMM 2023
Developer
- **Developed** a **web-based decision support platform** to **predict retail stockouts** and recommend targeted inter-store inventory transfers, minimizing **lost sales** across multi-location retail networks.
- **Created** an interactive dashboard using **React** and **TypeScript** to visualize **product demand trends** and real-time stock availability across multiple store locations.
- **Integrated Supabase** with **PostgreSQL** for real-time data storage, **user authentication**, and **REST API management**.
- **Implemented** a machine learning demand forecasting pipeline utilizing **Random Forest**, **Ridge**, and **Lasso regression**, achieving **R2 ≈ 0.97** on historical sales data.

Technologies / Tools Used : *React, TypeScript, Supabase, Python, Scikit-learn*

MedFL: Privacy-Preserving Federated Learning Platform | Github — MMM 2023
Architect
- Architected a **6-service containerized federated learning platform** enabling collaborative medical model training across federated client nodes while ensuring sensitive patient data remains decentralized.
- Implemented the **DWFed** (Dynamic Weight Federated) research algorithm for non-IID data aggregation, enhancing global model convergence under heterogeneous data distributions.
- Designed a **Docker Compose** microservices architecture, isolating the aggregation server, client nodes, data pipeline, and monitoring into independent, scalable services.
- Established a **REST API** communication layer to coordinate secure model weight exchanges across all 6 client-aggregator service boundaries, facilitating round-based training synchronization.

Technologies / Tools Used : *Python, PyTorch, Docker, Federated Learning*

SKILLS

Databases :	MongoDB, PostgreSQL, Supabase, Dexie.js
Frameworks Libraries :	React.js, Node.js, Express.js, Next.js, Vite
Programming Languages :	C++, Python, JavaScript, TypeScript, SQL, HTML5, CSS3
Tools Platforms :	Scikit-learn, PyTorch, Docker, Git, GitHub, Linux

AWARDS & ACHIEVEMENTS

- Selected among top 11 teams out of **7,000+** participants in a national innovation hackathon